FOR A FINITE MAP OF FINITE TYPE ALGEBRAS RELATIVE PSEUDO-COHERENCE IS DETECTED AFTER RESTRICTION OF SCALARS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The statement uses the letter m both for the integer in the pseudo-coherence bound and for the number of generators y_1, ..., y_m chosen in the proof; that coincidence is the author's text and is reproduced unchanged. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- Pseudo-coherence relative to a base ring and its behaviour under a finite pushforward. invoked at proof lines 21383-21389. Is the property being compared and the lemma that reduces it to the polynomial presentation.
- Finiteness of an algebra map giving monic equations for a generating set. used at proof lines 21371-21373. Produces the monic polynomials that present B as a quotient of a polynomial extension.

Source reference. Original lemma 16283: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16283–16283. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $A \to B$ be a finite map of finite type R-algebras. Let $m \in \mathbf{Z}$. Let $K^\bullet$ be a complex of B-modules. Then $K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent) relative to R if and only if $K^\bullet$ seen as a complex of A-modules is m-pseudo-coherent (pseudo-coherent) relative to R.*

Proof. Choose a surjection $R[x_1, \ldots, x_n] \to A$. Choose $y_1, \ldots, y_m \in B$ which generate B over A. As $A \to B$ is finite each y_i satisfies a monic equation with coefficients in A. Hence we can find monic polynomials $P_j(T) \in R[x_1, \ldots, x_n][T]$ such

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that $P_j(y_j) = 0$ in B. Then we get a commutative diagram

$$\begin{array}{ccc} & & R[x_1, \ldots, x_n, y_1, \ldots, y_m] \\ & & \downarrow \\ R[x_1, \ldots, x_n] & \longrightarrow & R[x_1, \ldots, x_n, y_1, \ldots, y_m]/(P_j(y_j)) \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array}$$

The top horizontal arrow and the top right vertical arrow satisfy the assumptions of Lemma 16283. Hence $K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent) as a complex of $R[x_1, \ldots, x_n]$-modules if and only if $K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent) as a complex of $R[x_1, \ldots, x_n, y_1, \ldots, y_m]$-modules. $\square$